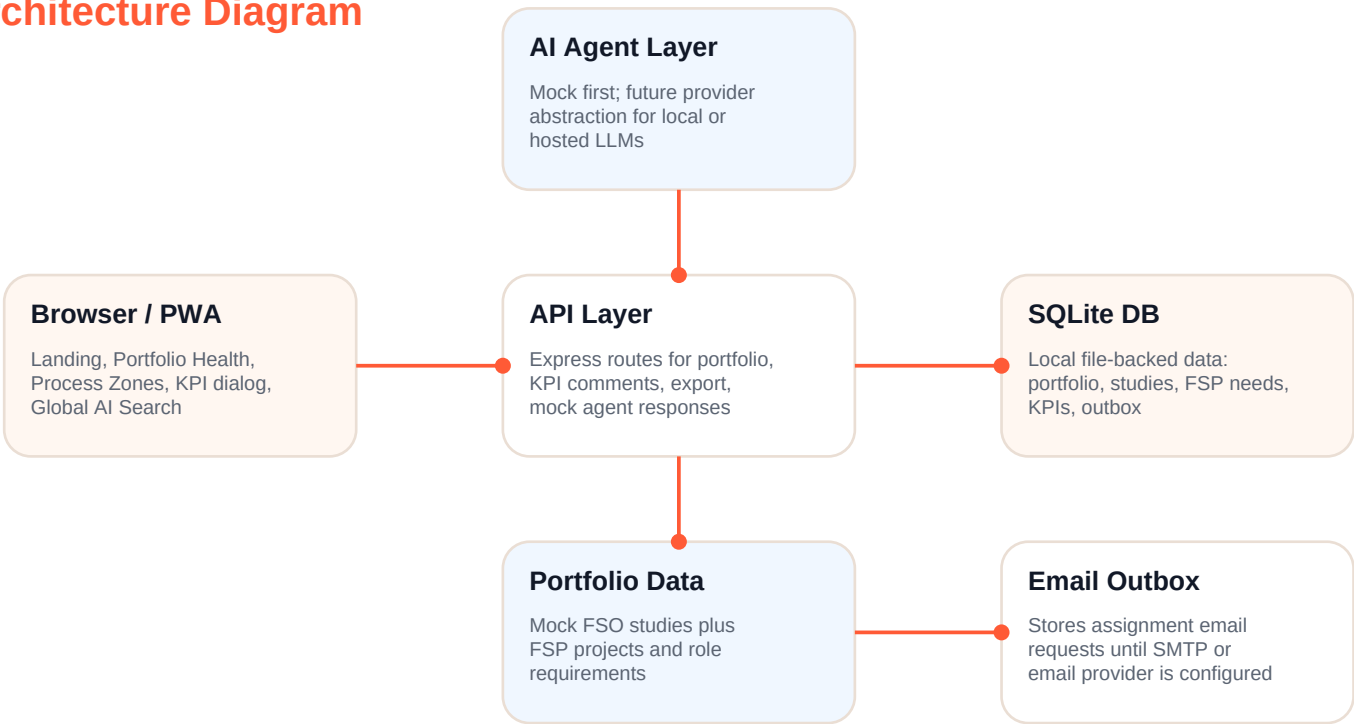


Clinical Portfolio Management Solutions

Architecture and Design Documentation

This document separates architecture and design guidance from the application UI. The prototype is local-first and uses mock portfolio data: FSO clinical studies plus FSP projects and role requirements.

Architecture Diagram



Core Data Flows

Flow	Description
Portfolio open	Browser loads portfolio, process-zone health, KPI metrics, FSO studies, and FSP requirements from the API.
Process-zone drill-down	User selects a process zone and sees metrics plus detailed records relevant to that zone.
KPI comment	User hovers over a KPI, opens the action dialog, and saves a comment against the metric.
Assign to	User enters an assignee email; the local prototype writes an email request to the email outbox.
AI search	Global search sends the user question to the API mock agent and returns a contextual answer.
Export	Portfolio KPIs, FSO studies, FSP requirements, KPI comments, and agile items export to XLSX.

Design System Notes

The application uses a separate modern SaaS identity rather than the strategy-deck visual system. The default theme is warm, minimal, grid-backed, and AI-forward, with dark mode support. The UI emphasizes dashboard density, process-zone drill-down, and action-oriented KPI interactions.

Data Model Summary

Entity	Purpose
Portfolio	Top-level operating unit formerly described as Engagement.
Process Zone	Scoping & Pricing, Contracts, Resource Management, Talent, People, Finance, Oversight.
KPI Metric	Metric value, target, trend, status, and process-zone association.
FSO Study	Mock clinical trial records currently in the portfolio.
FSP Requirement	Smaller projects or role requirements such as CRA, medical writing, safety, or data management support.
KPI Comment	User comment saved against a specific KPI metric.
Email Outbox	Local record of assignment emails pending real SMTP/provider setup.

Implementation Caveats

The first cut is local-only. The database is a file-backed SQLite database through sql.js. Email assignment currently writes to an outbox table because no SMTP or email provider credentials are configured. When the app moves beyond local prototype mode, the recommended path is to replace the local database with PostgreSQL, add a production auth provider, configure object storage, and connect email through SMTP or a transactional email service.